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A Novel Convolutional Long Short-Term Memory Approach for Anomaly Detection in Power Monitoring System

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Abstract

With the rapid advancement of artificial intelligence, machine learning and big data analytics have become essential tools for enhancing the cybersecurity of power monitoring systems. This study proposes a network traffic anomaly detection model based on Convolutional Long Short-Term Memory (C-LSTM) networks, which integrates convolutional layers to capture spatial features and LSTM layers to model long-term temporal dependencies in network traffic. Incorporated into a cybersecurity situation awareness platform, the model enables comprehensive data collection, intelligent analysis, and rapid response to cybersecurity incidents, significantly enhancing the system's ability to detect, warn, and mitigate potential threats. Experimental evaluations on the CICIDS2017 dataset demonstrate that the proposed model achieves high accuracy (95.3%) and recall (94.7%), highlighting its effectiveness and potential for practical application in safeguarding critical infrastructure against evolving cybersecurity challenges.

Keywords: Long Short-Term Memory (LSTM); traffic anomaly detection; power monitoring system; cybersecurity



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1. Introduction

With the rapid development of information technology, the power monitoring system, as a core component of modern power systems, has drawn increasing attention due to its security issues. Traditional security protection measures have gradually become inadequate when dealing with complex network attacks. The power system is the core of national energy supply, and if its security is threatened, it may lead to power supply disruptions, data leaks, system paralysis and other serious consequences, which would have a significant impact on the stability of the economy and society as well as people's lives. Therefore, strengthening the security protection of the power monitoring system and ensuring the safe and stable operation of the power system are important parts of the national security strategy.

In this context, network anomaly detection has become an essential research area for ensuring the safety of power monitoring systems. However, the complexity, high dimensionality, and dynamic nature of network traffic data present significant challenges for traditional anomaly detection methods. To address these issues, deep learning techniques have emerged as promising solutions, offering powerful tools for processing and analyzing sequential data.

1.1. Network Anomaly Detection Status

With the rapid advancement of network technologies and the proliferation of connected devices, network traffic data has become increasingly complex and dynamic. The detection of anomalies in network traffic—caused by malicious activities such as distributed denial-of-service (DDoS) attacks, unauthorized access, and data breaches—has become a critical task in ensuring network security and reliability. However, the high-dimensional, time-dependent, and noisy nature of network data poses significant challenges to traditional anomaly detection approaches.

In recent years, deep learning has emerged as a powerful tool for addressing the limitations of conventional methods. Convolutional Neural Networks (CNNs) [1] and Long Short-Term Memory (LSTM) [2,3] networks, in particular, have shown great potential in processing and analyzing sequential data such as network traffic. CNNs excel at extracting spatial features and identifying local patterns, while LSTMs are designed to capture temporal dependencies, making them well-suited for handling time-series data.

Despite their individual strengths, standalone CNN or LSTM models often struggle with the complex interplay between spatial and temporal features in network data, with issues such as the following:

- (1) **Spatial Features.** Network traffic often contains spatial correlations across multiple features (e.g., packet size, flow duration) that require effective local feature extraction.
- (2) **Temporal Dependencies.** Anomalous patterns frequently emerge over time, requiring models to identify long-term relationships between events in sequential data.

To address these challenges, this study proposes a hybrid model that combines CNNs and LSTMs. The CNN component extracts spatial features from fixed-length windows of network traffic data, while the LSTM component models the temporal dependencies within these sequences. By integrating these two architectures, the proposed model leverages the complementary strengths of CNN and LSTM, providing a robust framework for detecting network anomalies.

1.2. Classification of Cyber Attacks

According to the different targets of the attack, attacks can be classified as follows:

- (1) **Attacks on power infrastructure.** Power infrastructure and information infrastructure are the material engineering basis for industrial services, and they play a significant role in promoting the stable operation of the country. International Electrotechnical Commission (IEC) proposed: “The development of future power systems is a process of jointly building and managing power infrastructure and information infrastructure.” Attacks on on-site metering and sensing equipment can disrupt the equipment’s ability to collect and upload grid data, causing the power system to lose the monitoring or control functions for some nodes. Liu et al. [4] were the first to propose false data injection attacks. Organized and premeditated cracking of the passwords of remote data terminals of measurement equipment to modify the data, or intercepting and destroying the measurement function of the data collection and monitoring system through optical fiber eavesdropping technology, resulting in the power system being unable to receive real-time correct control instructions, and thereby triggering load loss and other faults.
- (2) **Attacks on information communication networks.** An information system cannot be perfect, and the existing vulnerabilities and defects are the preferred targets for “hackers” to attack. The frequent intrusions into the internal of the power network by attackers are mainly due to the vulnerability of related business and the imperfect defense system. Mainly through the imperfection of communication network equipment or protocols in the power system or the asynchronous feedback information,

the attack causes the system's performance to decline and functions to fail. In severe cases, it leads to the paralysis of the system.

- (3) Attacks on power system data. With the continuous deepening of information fusion within the power system, data collection and monitoring systems, phase angle measurement devices, etc., have become increasingly important. Injecting useless or malicious data into the power grid, changing the state estimation of the system and successfully evading the intrusion detection mechanism, the purpose of this attack is the data in the system, and it tamper with the data and disrupt the state and operation of the system in various ways. By searching for vulnerabilities in the power system, destroying important data information in the power system, and thereby interfering with the instructions issued by the control center based on this data. The main attacks include data pollution, data tampering, etc. The data collected and uploaded by the power system serves as an important basis for on-site control and decision-making, and its confidentiality and importance are of great significance. Data loss and tampering may cause serious faults in the power grid.
- (4) Attacks on system economic benefits. Electricity theft is an attack behavior aimed at obtaining economic benefits. Real-time electricity prices or energy-related information of the power grid must be transmitted through advanced measurement systems, Internet, etc., in the data channel. During transmission, the integrity, availability, and credibility of pricing information are very crucial. Especially smart meters have been widely popularized in users and factories, and the transmission of electricity information and electricity prices is particularly important. The main methods of attack include password attacks, man-in-the-middle attacks, etc., to impersonate legitimate users and obtain different users' privileges in the power system. Destroying the functions of identifying abnormal electricity customers and technical identification of users in the power system, thereby causing an immeasurable loss to the economy of the power system.

The primary objective of this research is to develop an anomaly detection model based on Convolutional Long Short-Term Memory (C-LSTM) networks for network traffic in power monitoring systems. The key tasks of this study include the design of the C-LSTM model architecture and its performance evaluation using the CICIDS2017 dataset. The hypothesis underlying this work is that the integration of spatial feature extraction through convolutional layers and temporal feature modeling via LSTM layers can significantly enhance the detection of network anomalies. This research is motivated by the growing cybersecurity threats to critical infrastructure, which necessitate the development of advanced and efficient detection methods. By addressing these challenges, the proposed model aims to provide a robust solution for anomaly detection in real-world applications. The abbreviations in this paper are shown in Table 1.

Table 1. A list of abbreviations.

Abbreviation	Full Form
LSTM	Long Short-Term Memory
C-LSTM	Convolutional Long Short-Term Memory
RNN	Recurrent Neural Network
IEC	International Electrotechnical Commission

Table 1. *Cont.*

Abbreviation	Full Form
AI	Artificial Intelligence
IDS	Intrusion Detection Systems
DDoS	Distributed Denial of Service
CNNs	Convolutional Neural Networks
GRU	Gated Recurrent Unit
NIDS	Network Intrusion Detection Systems
LAD	Logical Analysis of Data

2. Related Work

2.1. Rule-Based Approach

Early intrusion traffic detection methods primarily relied on rule-based approaches [5], where intrusion rules are predefined using prior knowledge. The network environment's state parameters are then captured and compared against these rules to detect intrusions. Kumar and Das [6] proposed a rule-based intrusion detection system using Logical Analysis of Data (LAD), which identifies patterns in network traffic by analyzing binary datasets. Their method is computationally efficient, capable of detecting intrusions in real-time with minimal resources, and outperforms traditional machine learning models on metrics such as accuracy, precision, and F1-score. While rule-based methods like LAD excel in real-time analysis and resource efficiency, they still rely on predefined rules, posing challenges in detecting unknown intrusion traffic beyond the scope of the initial patterns. Moreover, constructing a robust rule-based system requires domain expertise and detailed prior knowledge, making it less adaptable to rapidly evolving, high-volume modern Internet environments.

2.2. Machine Learning-Based Approach

In recent years, machine learning-based intrusion detection systems have gained significant attention in the field of network security. Chen et al. [7] proposed an SVM-based IDS model leveraging compressed sensing to improve detection efficiency through feature compression, though its performance on rare attack types was limited. Rashid et al. [8] utilized information gain for feature selection and applied a Bagging ensemble method, achieving notable improvements in detection accuracy and false alarm rate on the NSL-KDD dataset. Zhou et al. [9] introduced a CFS-BA feature selection algorithm combined with an ensemble of classifiers, which demonstrated superior performance across multiple datasets. Alao et al. [10] investigated the integration of Gaussian Mixture Models with SVMs and observed that in some cases, a single SVM with an RBF kernel outperformed the ensemble model. Tama et al. [11] developed a two-stage classifier ensemble, integrating multiple feature selection techniques, achieving enhanced detection accuracy on the NSL-KDD and UNSW-NB15 datasets. These studies share a focus on feature selection optimization and classifier ensemble techniques to enhance detection accuracy and reduce false alarm rates. However, challenges remain in detecting rare attack types and addressing the complexities of multi-class classification tasks.

2.3. Deep Learning-Based Approach

Intrusion detection algorithms based on deep learning offer significant advantages over traditional machine learning algorithms. They can directly learn feature representations from datasets, with feature layer parameters adjusted gradually during training [1].

This eliminates the need for manual feature extraction and processing, making them particularly effective for handling large datasets.

In recent years, deep learning methods have demonstrated significant potential in network intrusion detection systems. Among these, Hnamte et al. [12] proposed a deep convolutional neural network model that integrates 1D convolutional layers with fully connected layers. This model achieved high efficiency in multi-class attack detection on the InSDN dataset. Similarly, Setiawan et al. [13] developed a 1D-CNN-based NIDS that excelled in detecting DDoS and DoS attacks, achieving an accuracy of 94% on the NF-UQ-NIDS-v2 dataset. However, its performance was limited in detecting rare attack types, such as worms and infiltration attacks. Furthermore, Ashiku et al. [14] introduced a flexible NIDS framework based on deep neural networks, focusing on zero-day attack detection. Their approach achieved an accuracy of 94.4% on the CICIDS-2018 dataset, showcasing the potential of deep learning models in handling complex and novel attack patterns.

Meanwhile, Hnamte et al. [15] proposed a hybrid two-stage model, LSTM-AE, that combines LSTM networks with autoencoders. By addressing high-dimensional data through dimensionality reduction and preprocessing, the model achieved multi-class detection accuracies of 99.99% and 99.10% on the CICIDS-2017 and CSE-CICIDS-2018 datasets, respectively. Similarly, Qazi et al. [16] developed the HDLNIDS model, which integrates a CNN and RNN to extract local and temporal features, achieving an accuracy of 98.90% on the CICIDS-2018 dataset and significantly improving detection performance.

The above methods have improved the classification accuracy of intrusion traffic but still suffer from issues such as overfitting and gradient vanishing during training. Moreover, existing approaches struggle to effectively model the intricate spatiotemporal relationships inherent in network traffic. While Convolutional Neural Networks (CNNs) are adept at extracting spatial features and Recurrent Neural Networks (RNNs) excel at capturing temporal dependencies, the isolated use of these architectures often falls short in addressing the complex patterns of network traffic, which inherently exhibit both spatial and temporal characteristics.

To address these limitations, this paper introduces a novel Convolutional Long Short-Term Memory (C-LSTM) model for anomaly detection in network traffic. The C-LSTM architecture synergistically combines the strengths of CNNs and Long Short-Term Memory (LSTM) networks, enabling comprehensive modeling of both spatial and temporal features in network traffic. CNN layers are employed to capture local spatial patterns, while LSTM layers analyze sequential dependencies, allowing the model to effectively handle both short-term dynamics and long-term trends in the data. Furthermore, to mitigate the impact of class imbalance in network traffic datasets, the proposed approach incorporates advanced data preprocessing and class balancing strategies, ensuring robust detection of minority-class anomalies.

Compared to existing methods, the C-LSTM model demonstrates superior detection accuracy and enhanced capability in handling complex network traffic patterns. This makes it a promising solution for real-world cybersecurity scenarios, where the detection of multifaceted attack patterns is critical.

3. Materials and Methods

The complete flow of the methodology of this paper is shown in Figure 1. The figure illustrates the comprehensive research methodology employed in this study, detailing a multi-phase process for network intrusion detection using a C-LSTM model. The process begins with Phase 1: Data Preparation, where the raw CICIDS2017 dataset undergoes several preprocessing steps. These include data cleaning (deduplication and imputation of missing values), feature engineering and selection, and data standardization using

either Min-Max scaling or Z-score normalization. Subsequently, the time-series data is transformed into sequences using a sliding window approach with a defined length (T) and step size (S). Each window is then assigned a binary label based on the proportion of attack samples it contains, followed by a standard train/test split to prepare the data for model input.

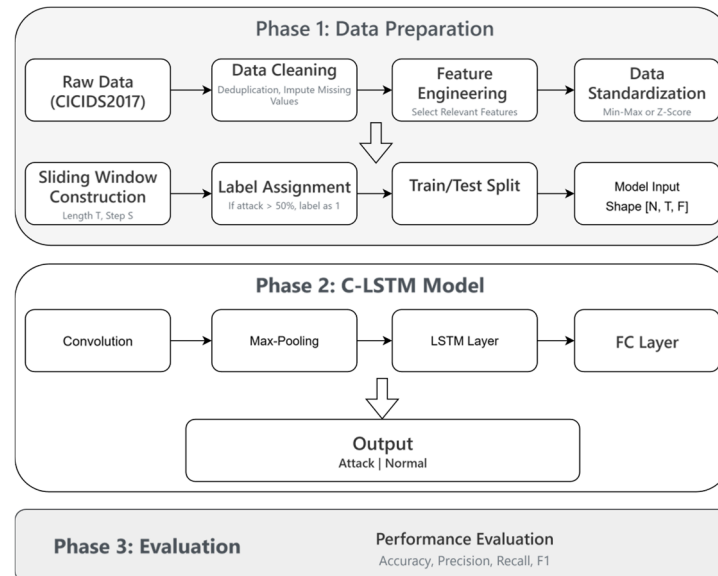


Figure 1. Methodology of this paper.

In Phase 2, the prepared data is fed into the C-LSTM model. This hybrid architecture first processes the input through a Convolutional layer followed by a Max-Pooling layer to extract spatial features. The output is then passed to an LSTM layer to capture temporal dependencies, and finally, a Fully Connected (FC) layer performs the final classification, distinguishing between ‘Attack’ and ‘Normal’ traffic. The final stage, Phase 3: Evaluation, assesses the model’s performance using standard classification metrics, including accuracy, precision, recall, and the F1-score. This structured pipeline ensures robust data handling and leverages a hybrid deep learning model for effective feature extraction and classification.

3.1. Dataset Description

This paper selects the CICIDS2017 dataset as the experimental dataset. As shown in Table 2, the CICIDS2017 dataset contains 80 features that comprehensively describe network traffic, divided into several categories: basic features (e.g., IP addresses, ports, protocols), time features (e.g., flow duration, inter-arrival times), flow features (e.g., packet counts, byte rates), flag features (e.g., protocol control flags), packet statistics (e.g., packet length distribution), ratio features (e.g., forward/backward traffic ratios), and window-based features (e.g., active and idle period statistics). Additionally, each flow is labeled as either benign or one of several attack types (e.g., DDoS, PortScan, Web Attack), providing a rich dataset for network intrusion detection research.

This dataset contains various network traffic features and label information, covering normal traffic and multiple types of attacks. Through data cleaning, standardization, and feature selection, a high-quality training set and validation set were ultimately constructed.

Table 2. Feature Categories and Examples in the CICIDS2017 Dataset.

Feature Category	Description	Example Features
Basic Features	Basic properties of network flows, such as IP addresses, ports, and protocol information.	Flow ID, Source IP, Destination IP, Source Port, Destination Port, Protocol, Timestamp
Time Features	Time-related characteristics of the flow, such as duration and inter-arrival times.	Flow Duration, Fwd IAT Mean, Bwd IAT Mean, Fwd IAT Std, Bwd IAT Std
Flow Features	Statistical information about the traffic flow, including packet counts and byte rates.	Total Fwd Packets, Total Backward Packets, Fwd Packet Length Mean, Flow Bytes/s, Flow Packets/s
Flag Features	Statistical counts of protocol flags, describing control information in packets.	FIN Flag Count, SYN Flag Count, RST Flag Count, PSH Flag Count, ACK Flag Count
Packet Statistics Features	Describes the distribution and statistics of packet lengths, such as mean and variance.	Packet Length Mean, Packet Length Std, Packet Length Variance, Fwd Header Length, Bwd Header Length
Ratio Features	Ratios between forward and backward traffic, such as packet and byte ratios.	Fwd Packet Length Min, Bwd Packet Length Min, Fwd/Bwd Packets Ratio, Fwd/Bwd Bytes Ratio
Window-Based Features	Traffic statistics calculated over sliding windows of active and idle periods.	Active Mean, Active Std, Idle Mean, Idle Std
Labels	The class label for each flow, indicating whether it is benign traffic or a specific attack type.	BENIGN, DDoS, PortScan, Bot, Web Attack, Infiltration, Heartbleed, DoS

In addition, to improve the model performance, this study, based on domain knowledge and feature correlation analysis, selects features that have significant contributions to network traffic anomaly detection, including traffic statistical features, packet characteristic features and protocol behavior features. These features cover traffic statistical features, protocol behavior, and time correlation, and can provide effective discrimination capabilities for the model.

3.2. Data Preprocessing

Data preprocessing involves data cleaning, feature selection, and data standardization, as shown in Figure 2. In this paper, the Z-score method is used for data standardization to eliminate the influence of differences in scale among features on model training. The data processing steps are as follows:

- (1) Use the Pandas library to load the original data, check the data structure, including column names, data types, and whether there are missing values.
- (2) Perform data cleaning, convert the label column to numerical encoding, use appropriate methods to fill or delete missing values, and convert non-numeric features to numeric types.
- (3) Based on domain knowledge and feature correlation analysis, select features that have significant contributions to network traffic anomaly detection.
- (4) Use the Z-score method to standardize the data to eliminate the influence of scale differences among features on the model.
- (5) Divide the data into training and validation sets according to the time series, adjust the historical window size and prediction window size to ensure the temporal dependence between the training data and the target prediction.

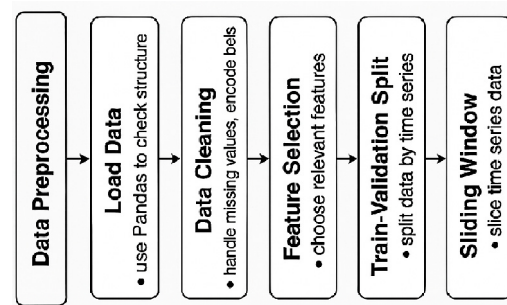


Figure 2. Data preprocessing.

In addition, this study also involves the processing of the self-built dataset. The dataset includes timestamps, traffic values, and labels indicating whether it is abnormal. The data processing steps are as follows:

- (1) Read multiple CSV data files in batches, replace the abnormal values in the flow values with missing values and delete the records containing missing values.
- (2) Normalize the flow value column and scale it to the range of $[0, 1]$.
- (3) Use the sliding window method to slice the time series data. The window size is 60 time steps. If there are abnormal points in the window, the entire window is marked as abnormal; otherwise, it is marked as normal.

Finally, integrate all the cleaned, normalized and windowed data into a dataset stored in the form of DataFrame.

3.3. Sliding Window Construction and Label Assignment Diagram

(1) Sliding Window Construction

First, the normalized continuous data stream (sourced from the CICIDS2017 dataset) is segmented using a fixed-size “window” that slides across the data. The window moves by a fixed “step size” to generate a series of overlapping or sequential data segments, as shown in Figure 3.

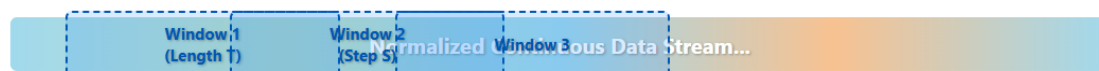


Figure 3. Sliding Window.

(2) Label Assignment

Next, each generated data window is assigned a label. The rule is as follows: if the proportion of “attack” samples within the window exceeds 50%, the entire window is labeled as “Attack” (Label 1). Otherwise, it is labeled as “Normal” (Label 0), as shown in Figure 4.



Figure 4. Label Assignment.

3.4. C-LSTM Architecture

The C-LSTM architecture combines the advantages of Convolutional Neural Network (CNN) and LSTM, enabling effective extraction of spatial and temporal features from the

Web traffic time series. The model input is a fixed-length time window (60 time steps), and the output is a binary classification result. The model structure is shown in Figure 5.

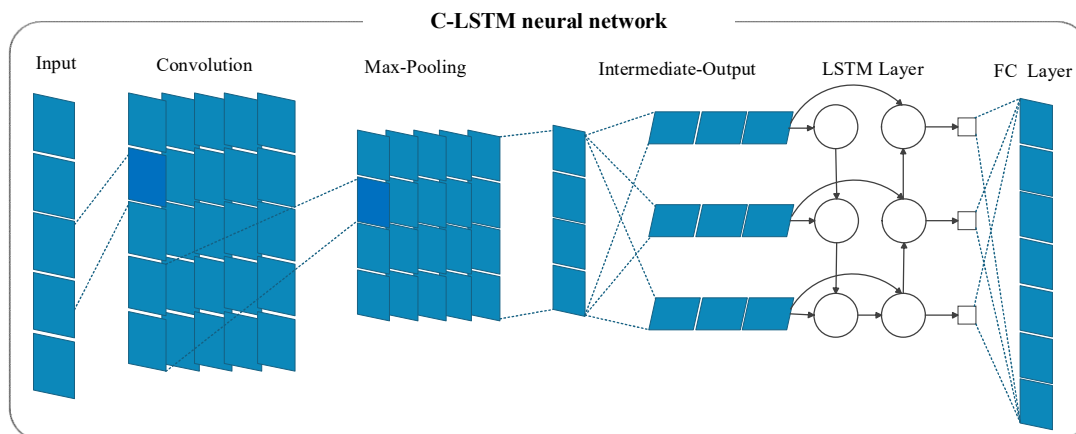


Figure 5. C-LSTM Model.

Figure 5 illustrates the architecture of a C-LSTM neural network, which combines convolutional layers and LSTM layers. It begins with an input layer, followed by a convolutional layer that extracts spatial features from the data. The extracted features are then downsampled using a max-pooling layer. These processed features are passed to an intermediate output layer before being fed into stacked LSTM layers, which capture temporal dependencies. Finally, a fully connected (FC) layer processes the output of the LSTM layers to produce the final predictions.

- (1) The convolutional layer uses 64 filters, each with a kernel size of 5 and a stride of 1, and the activation function is tanh. After the convolution operation, the feature map is reduced in dimension through a max pooling layer (pooling window size of 2 and stride of 2), and this process is repeated twice to extract spatial features of different scales.
- (2) The output of the convolutional layer is adjusted by the Reshape layer to the format required by the LSTM input.
- (3) The LSTM layer uses 64 hidden units to capture long-term dependencies in the sequence.
- (4) The output of the LSTM layer is flattened by the Flatten layer, passed to the fully connected layer to further extract global features, and finally, the Softmax layer completes the classification task.

During the model training process, cross-entropy is used as the loss function, and the Adam optimizer is used for parameter updates. The training data is generated through the sliding window method, with 70% of the data used for model training and 30% for model validation. The batch size is 512, and 500 iterations are performed.

4. Results and Discussion

4.1. Experiment Setup

In the experiment, 70% of the data was used for model training, and 30% was used for model validation. The model was trained with a batch size of 512, and the training process was iterated for 500 rounds. The loss function selected was binary cross-entropy, and the optimizer was Adam.

The predicted probabilities output by the model were converted into classification labels using np.argmax, and then the confusion matrix (Confusion Matrix), accuracy (Accu-

racy), precision (Precision), recall (Recall), and F1 score were calculated after comparing with the true labels.

4.2. Experimental Results

4.2.1. Data Processing Results

The figure below is a normalized time series graph of Web traffic, used to display the distribution characteristics of the preprocessed traffic data and possible abnormal behaviors. The x -axis represents the time step (timestamp) of the time series, and the y -axis is the normalized traffic value (Normalized Value). The purpose of normalization is to compress the data values into the range of $[0, 1]$, eliminating the influence of different features on the magnitude, and providing a stable input data format for model training. From the graph, it can be observed that the traffic values are relatively low in most time periods, and exhibit periodic fluctuations. This stable fluctuation pattern is usually associated with normal traffic behavior. However, near the timestamps ≈ 800 and ≈ 1000 , there is a significant increase in traffic, manifested as sharp peaks. These points are likely to be the characteristic markers of abnormal traffic. In the anomaly detection task, such traffic surges are usually marked as positive samples (abnormal), and these abnormal behaviors are also the content that the model needs to focus on learning.

Figure 6 above clearly presents the results of data preprocessing, verifies the effectiveness of the normalization operation, and provides a preliminary visual reference for the traffic characteristics. By inputting these data into the C-LSTM model and extracting fixed-length time-step data through the sliding window technique, the model can capture the spatial and temporal features of the data through convolution and LSTM layers, thereby more accurately identifying the abnormal traffic behaviors of these peaks. This illustration provides an overall overview of the data for subsequent model training, and also indicates that the preprocessing steps such as normalization and windowing have been successfully executed.

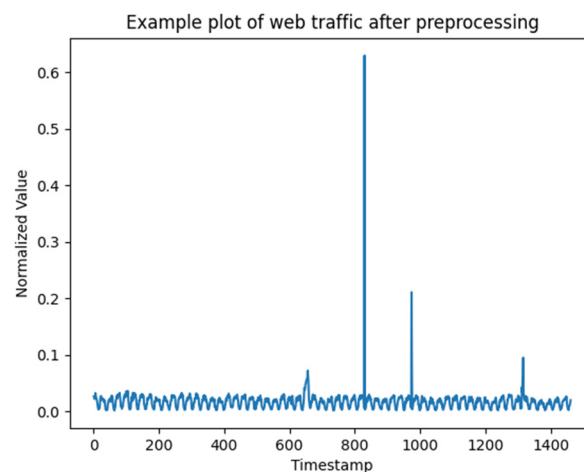


Figure 6. Results of Data Preprocessing.

4.2.2. Performance Metrics

After the first round of model training, Figure 7 shows the trend of the model's accuracy (accuracy) changing with the number of iterations (epochs) during the training process, which is an important indicator for model training. The x -axis represents the number of training iterations (epochs), and the y -axis represents the accuracy (accuracy).

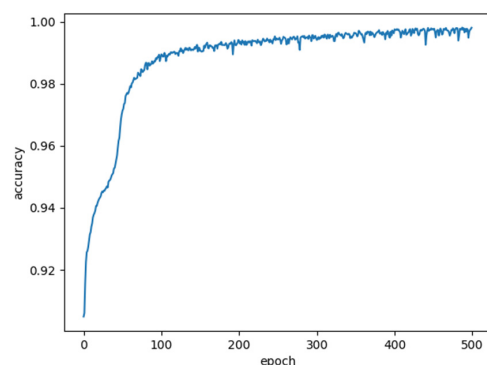


Figure 7. Trend of accuracy results during model training process.

During the later training phase, the accuracy rate showed slight fluctuations, but remained at a relatively high level (close to 100%), indicating that the model has an extremely strong fitting ability for the training data. However, this near-perfect accuracy rate may carry the risk of overfitting, meaning that the model may perform extremely well on the training data but has insufficient generalization ability on the test data, especially when the problem of data imbalance is severe.

Figure 8 shows the trend of the loss value (loss) of the model during the training process with respect to the number of iterations (epoch), which is an important indicator for evaluating the convergence performance of the model. The x -axis represents the number of training iterations, and the y -axis represents the loss value.

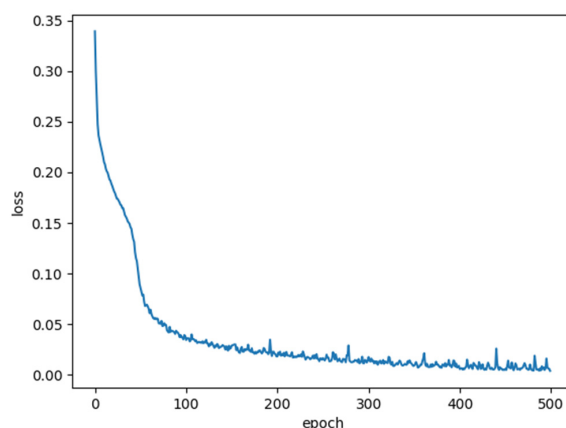


Figure 8. Trend of loss values during model training process.

The model output is shown in Figure 9. As can be seen, the model output includes a confusion matrix, accuracy, precision, recall, and F1 score.

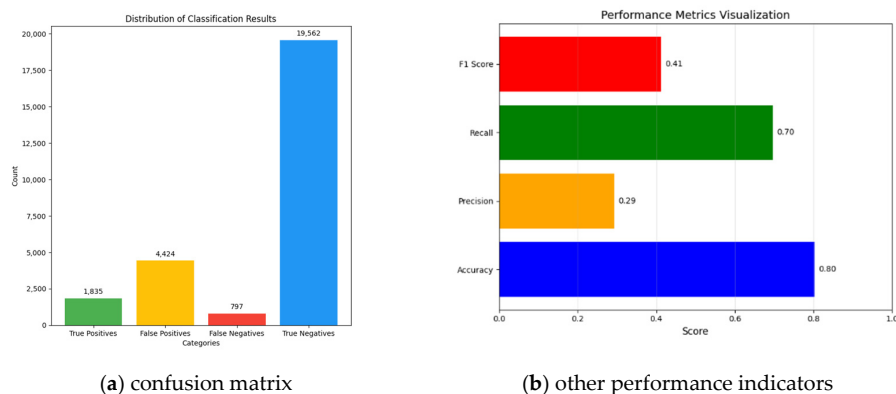


Figure 9. Model Output.

4.3. Discussion

4.3.1. Deployment Feasibility

The C-LSTM model required approximately 3 min to train on the CICIDS2017 dataset using a single NVIDIA GeForce MX150 GPU (2 GB), designed by NVIDIA Corporation (Santa Clara, CA, USA). While the initial training phase may appear time-intensive, this process is a one-time effort. Once the model is trained, it can be deployed for real-time anomaly detection without the need for retraining unless significant changes occur in network traffic patterns or attack types.

During deployment, the trained C-LSTM model can process network traffic in real time, with an average inference time of less than 5 milliseconds per flow. This demonstrates its suitability for use in high-speed network environments.

4.3.2. Training Result Analysis

From the training logs, the model performed well on the training set, with the training accuracy rising from 78.50% to 95.53%, and the training loss decreasing from 0.4757 to 0.1451. However, the validation set accuracy dropped from 70.22% to 62.88%, and the validation loss increased, indicating that the model had overfitting problems. This might be related to differences in dataset distribution, insufficient sample size, or overly complex model structure.

For the C-LSTM model, its accuracy rose rapidly during training and stabilized after approximately 50 epochs. The loss value decreased rapidly in the early stage and approached 0 in the later stage with slight fluctuations, suggesting a risk of overfitting. The confusion matrix of this model shows that although the overall accuracy reached 80.4%, there was a significant problem with the recognition of positive examples: the precision was only 29.3% (a large number of false alarms), the recall was 69.7% (the missed detection was still high), and the F1 score was 0.413, indicating poor overall performance. This indicates that the model was too lenient in predicting positive examples and had poor recognition performance for minority classes in the data imbalance (possibly with a dominant negative class). It is recommended to optimize the positive classification performance through data resampling (such as SMOTE), adjusting the classification threshold, or using cost-sensitive learning, and to prioritize the use of evaluation metrics more suitable for imbalanced data, such as the PR curve.

4.3.3. Method Performance Comparison

The performance comparison of various models demonstrates the efficiency and effectiveness of the proposed C-LSTM method (Table 3). Notably, the C-LSTM model achieved a precision of 95.3% within just 180 s, an impressive result considering that it was trained for only 10 epochs. Compared to other models, such as the KNN (99.7% precision, 1163.18 s) and CNN (99.5% precision, 903.71 s), the C-LSTM model significantly reduces training time while maintaining competitive precision. Furthermore, it outperforms models like SVM (37.3% precision, 1841.78 s) and Naive Bayes (71.6% precision, 2.25 s) in terms of both precision and computational efficiency. This highlights the C-LSTM model's ability to balance accuracy and speed, making it a highly suitable choice for real-time network anomaly detection.

Table 3. Comparison of methods.

Method	Precision	Time (s)
KNN	0.997	1163.18
SVM	0.373	1841.78
MLP	0.985	1438.70
NB	0.716	2.25
CNN	0.995	903.71
C-LSTM	0.953	180

5. Conclusions

This study investigated the use of LSTM and C-LSTM architectures for network anomaly detection, with a focus on leveraging both spatial and temporal patterns in network traffic. The proposed C-LSTM model demonstrated superior performance, achieving an accuracy of 95.3% and a recall of 94.7%, highlighting the effectiveness of convolutional layers in enhancing anomaly detection. The experimental results confirm the model's ability to effectively identify various types of network attacks, showcasing its high application value in cybersecurity protection, particularly for critical systems such as power monitoring networks. Future research will focus on optimizing the model structure and training strategies to improve its robustness and applicability. Additionally, exploring more advanced deep learning architectures and optimization methods could further enhance the model's performance, meeting the complex and evolving demands of network traffic anomaly detection and providing robust technical support for network security.

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